

host_agent

Here is the result for your query:

- All buildings within 1 km and 2 km of the University of Exeter in Exeter have been identified and plotted.
- The map shows:
- The University of Exeter location(s)
- Two rings: one at 1 km and one at 2 km from the university
- Buildings within 1 km (highlighted in red, 15,238 buildings)
- Buildings within 2 km (highlighted in blue, 24,693 buildings)

If you would like to view or download the interactive map, let me know!

planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are buildings, and we need to find all buildings within 1 km and 2 km of the University of Exeter, which is also located in Exeter, so there will be 1 entry for the University of Exeter. The conditions specify the distances of 1 km and 2 km from the university.

output steps: ["Find Exeter 1 area", "Find University of Exeter in Exeter 1 result", "Find all buildings in Exeter as many search results", "Apply condition for buildings within 1 km of University of Exeter", "Apply condition for buildings within 2 km of University of Exeter"]

named_area_agent

The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) for Exeter, ensuring high accuracy by filtering using name and description columns.

No specific polygons for the University of Exeter in Exeter were found in the provided data. If you need further assistance or a different query, feel free to ask!

data_analysis_agent

Filtered to 1 polygon for Exeter. The results have been saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) for Exeter, filtered using name and description columns for high accuracy.

No polygons specifically matching the University of Exeter in Exeter were found in the provided artifact. There are no results to display for this filter.

Filtered locations for 'University of Exeter'. Number of entries: 22. An artifact has been generated with the name 'filtered_university_of_exeter_locations' and contains a GeoDataFrame with only the point locations specifically for 'University of Exeter', filtered using multiple relevant columns.

address_agent

The specific point locations for the "University of Exeter" have been filtered and saved. The artifact name is `filtered_university_of_exeter_locations`, containing the relevant data.

buildings_agent

I found the following artifacts for buildings within the 'filtered_exeter_area_polygon':

- 1. **buildings_exeter_buildings**: A geopandas dataframe containing building data. (Records: 69,169)
- 2. **buildingline_exeter_buildings**: A geopandas dataframe containing building line data. (Records: 69,342)
- 3. **buildingpart_exeter_buildings**: A geopandas dataframe containing building part data. (Records: 85,480)

These artifacts contain the building data you requested.

plotting_agent

['Plotted buildings within 1km (red, 15238) and 2km (blue, 24693) of university locations. First 5 buildings within 1km: [\'Mid-Terrace House\', \'Mid-Terrace House\', \'Unknown Building\', \'Ancillary Building\', \'Unknown Building\'], 'buildings_within_1km_2km_map', 'Folium map showing university locations, 1km and 2km buffer rings, and buildings within each ring.', 'buildings_within_1km_2km_university.html']

Tool Calls and Code

Tool: send_message

- Args: {'target': 'planning_agent', 'task_description': 'User wants to see all buildings within 1 and 2 km of University of Exeter in Exeter. Provide the general steps to solve this geospatial query.'}
- Response: The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are buildings, and we need to find all buildings within 1 km and 2 km of the University of Exeter, which is also located in Exeter, so there will be 1 entry for the University of Exeter. The conditions specify the distances of 1 km and 2 km from the university.

output steps: ["Find Exeter 1 area", "Find University of Exeter in Exeter 1 result", "Find all buildings in Exeter as many search results", "Apply condition for buildings within 1 km of University of Exeter", "Apply condition for buildings within 2 km of University of Exeter"]

Tool: send_message

- Args: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}
- Response: The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) for Exeter, ensuring high accuracy by filtering using name and description columns.

Tool: call_os_ngd

- Args: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}
- Response: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..., count of records fetched : 98.

Tool: send_message

- Args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}
- Response: Filtered to 1 polygon for Exeter. The results have been saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) for Exeter, filtered using name and description columns for high accuracy.

Tool: generate_metadata_for_artifacts

- Args: {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter using multiple columns to ensure we get the correct polygon for Exeter
    filtered = gdf[(gdf['name1_text'].str.lower() == 'exeter') |
                  (gdf['description'].str.contains('Exeter', case=False, na=False)) |
                  (gdf['theme'].str.contains('exeter', case=False, na=False))]
    # Further refine: ensure the most relevant polygon is selected if multiple
    if len(filtered) > 1:
        # Prefer exact match in name1_text
        filtered = filtered[filtered['name1_text'].str.lower() == 'exeter']
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
    return [summary, 'filtered_exeter_area_polygon', 'GeoDataFrame containing only the polygon(s) for Exeter, filtered using name and description columns for h
```

Tool: send_message

- Args: {'target': 'named_area', 'task_description': 'Find the area polygon for University of Exeter in Exeter.'}
- Response: No specific polygons for the University of Exeter in Exeter were found in the provided data. If you need further assistance or a different query, feel free to ask!

Tool: call_os_ngd

- Args: {'filters': ['University of Exeter'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'university_of_exeter_area_polygon'}
- Response: Search results have been found. Artifact generated is : university_of_exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for University of Exeter in Exeter..., count of records fetched : 72.

Tool: send_message

- Args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'university_of_exeter_area_polygon' to find the specific polygon for the University of Exeter in Exeter. Save the filtered results as 'filtered_university_of_exeter_area_polygon'."}
- Response: No polygons specifically matching the University of Exeter in Exeter were found in the provided artifact. There are no results to display for this filter.

Tool: generate_metadata_for_artifacts

- Args: {'artifact_names': ['university_of_exeter_area_polygon']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_university_of_exeter_polygon(data: list):
    gdf = data[0]
    # Filter using multiple columns for high accuracy
    mask = (
        (gdf['name1_text'].str.contains('Exeter', case=False, na=False)) &
        (gdf['description'].str.contains('University', case=False, na=False))
    )
    filtered = gdf[mask]
    summary = f"Filtered to {len(filtered)} polygon(s) for the University of Exeter in Exeter."
    artifact_name = 'filtered_university_of_exeter_area_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) for the University of Exeter in Exeter, filtered using name and description columns for'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool: send_message

- Args: {'target': 'address', 'task_description': "Find the point location for University of Exeter within the 'filtered_exeter_area_polygon'."}
- Response: The specific point locations for the "University of Exeter" have been filtered and saved. The artifact name is `filtered_university_of_exeter_locations`, containing the relevant data.

Tool: call_os_ngd

- Args: {'filters': ['University'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': False, 'street_address': False, 'filename': 'university_of_exeter_locations'}
- Response: Search results have been found. Artifact generated is : `university_of_exeter_locations`, Description is : A geopandas dataframe containing address data with filters and bbox applied for the query Find the point location for University of Exeter within the 'filtered_exeter_area_polygon'. using

filters ['University']. Further Name filtering is available for this, count of records fetched : 41.

Tool: send_message

- Args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'university_of_exeter_locations' to find the specific point location for 'University of Exeter'. Save the filtered results as 'filtered_university_of_exeter_locations'."}
- Response: Filtered locations for 'University of Exeter'. Number of entries: 22. An artifact has been generated with the name 'filtered_university_of_exeter_locations' and contains a GeoDataFrame with only the point locations specifically for 'University of Exeter', filtered using multiple relevant columns.

Tool: generate_metadata_for_artifacts

- Args: {'artifact_names': ['university_of_exeter_locations']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_university_of_exeter_location(data: list):
    gdf = data[0]
    # Filter for rows where the organisationname or subname or name contains 'University of Exeter'
    filtered = gdf[(gdf['organisationname'].str.contains('University of Exeter', case=False, na=False)) |
                  (gdf['subname'].str.contains('University of Exeter', case=False, na=False)) |
                  (gdf['name'].str.contains('University of Exeter', case=False, na=False)) |
                  (gdf['fulladdress'].str.contains('University of Exeter', case=False, na=False))]
    summary = f"Filtered locations for 'University of Exeter'. Number of entries: {len(filtered)}."
    artifact_name = 'filtered_university_of_exeter_locations'
    artifact_description = "A GeoDataFrame containing only the point locations specifically for 'University of Exeter', filtered using multiple relevant columns."
    return [summary, artifact_name, artifact_description, filtered]
```

Tool: send_message

- Args: {'target': 'buildings', 'task_description': "Find all buildings within the 'filtered_exeter_area_polygon'."}

Response: I found the following artifacts for buildings within the 'filtered_exeter_area_polygon':

buildings_exeter_buildings: A geopandas dataframe containing building data. (Records: 69,169)

- **buildingline_exeter_buildings:** A geopandas dataframe containing building line data. (Records: 69,342)

- **buildingpart_exeter_buildings**: A geopandas dataframe containing building part data. (Records: 85,480)

These artifacts contain the building data you requested.

Tool: call_os_ngd

- Args: {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'exeter_buildings'}
- Response: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['buildings_exeter_buildings', 'buildingline_exeter_buildings', 'buildingpart_exeter_buildings'], Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Find all buildings within the 'filtered_exeter_area_polygon'. using filters None.", ...], counts of records fetched are : [69169, 69342, 85480].

Tool: send_message

- Args: {'target': 'plotting_agent', 'task_description': "Plot all buildings from 'buildings_exeter_buildings' that are within 1 km and 2 km of the point(s) in 'filtered_university_of_exeter_locations'. Show two rings: one for 1 km and one for 2 km, and highlight the buildings within each ring."}
- Response: ['Plotted buildings within 1km (red, 15238) and 2km (blue, 24693) of university locations. First 5 buildings within 1km: ['Mid-Terrace House', 'Mid-Terrace House', 'Unknown Building', 'Ancillary Building', 'Unknown Building'], 'buildings_within_1km_2km_map', 'Folium map showing university locations, 1km and 2km buffer rings, and buildings within each ring.', 'buildings_within_1km_2km_university.html']

Tool: generate_metadata_for_artifacts

- Args: {'artifact_names': ['buildings_exeter_buildings', 'filtered_university_of_exeter_locations']}

Tool Code Executor

```
def plot_buildings_within_rings(data: list):  
    import geopandas as gpd  
    import folium  
    from shapely.geometry import Point  
    import matplotlib.pyplot as plt  
  
    # Unpack data  
    buildings = data[0]  
    uni_points = data[1]  
  
    # Ensure all CRS are the same and projected for distance calculation  
    # Use British National Grid (EPSG:27700) for buffer, then convert to EPSG:4326 for folium
```

```

buildings = buildings.to_crs(epsg=27700)
uni_points = uni_points.to_crs(epsg=27700)

# Create 1km and 2km buffers around all university points
buffer_1km = uni_points.buffer(1000)
buffer_2km = uni_points.buffer(2000)
# Union the buffers in case of multiple points
buffer_1km_union = buffer_1km.unary_union
buffer_2km_union = buffer_2km.unary_union

# Find buildings within 2km, and within 1km
buildings_within_2km = buildings[buildings.intersects(buffer_2km_union)]
buildings_within_1km = buildings[buildings.intersects(buffer_1km_union)]

# For plotting, convert everything to EPSG:4326
buildings_within_2km = buildings_within_2km.to_crs(epsg=4326)
buildings_within_1km = buildings_within_1km.to_crs(epsg=4326)
buildings = buildings.to_crs(epsg=4326)
uni_points = uni_points.to_crs(epsg=4326)

# Convert buffers to GeoSeries for plotting
buffer_1km_gdf = gpd.GeoSeries([buffer_1km_union], crs=27700).to_crs(epsg=4326)
buffer_2km_gdf = gpd.GeoSeries([buffer_2km_union], crs=27700).to_crs(epsg=4326)

# Create folium map centered on the mean of university points
center_lat = uni_points.geometry.y.mean()
center_lon = uni_points.geometry.x.mean()
m = folium.Map(location=[center_lat, center_lon], zoom_start=13)

# Plot 2km buffer (outer ring)
folium.GeoJson(buffer_2km_gdf.iloc[0], name='2km Buffer', style_function=lambda x: {'color': 'blue', 'fill': False, 'weight': 2, 'dashArray': '10,10'}).add_to(m)
# Plot 1km buffer (inner ring)
folium.GeoJson(buffer_1km_gdf.iloc[0], name='1km Buffer', style_function=lambda x: {'color': 'red', 'fill': False, 'weight': 2}).add_to(m)

# Plot university points
for idx, row in uni_points.iterrows():
    folium.Marker(location=[row.geometry.y, row.geometry.x], popup=row.get('description', 'University Location'), icon=folium.Icon(color='green', icon='university')).add_to(m)

# Plot buildings within 1km (highlighted)
folium.GeoJson(buildings_within_1km.geometry, name='Buildings within 1km', style_function=lambda x: {'color': 'red', 'fillColor': 'red', 'fillOpacity': 0.3}).add_to(m)
# Plot buildings within 2km but outside 1km (highlighted differently)
buildings_within_2km_only = buildings_within_2km[~buildings_within_2km.index.isin(buildings_within_1km.index)]
folium.GeoJson(buildings_within_2km_only.geometry, name='Buildings within 2km', style_function=lambda x: {'color': 'blue', 'fillColor': 'blue', 'fillOpacity': 0.3}).add_to(m)

folium.LayerControl().add_to(m)

# Save map
map_filename = 'buildings_within_1km_2km_university.html'

```



```
m.save(map_filename)
```

```
# Prepare summary
```

```
summary = f"Plotted buildings within 1km (red, {len(buildings_within_1km)}) and 2km (blue, {len(buildings_within_2km_only)}) of university locations. First
```

```
artifact_name = 'buildings_within_1km_2km_map'
```

```
artifact_description = 'Folium map showing university locations, 1km and 2km buffer rings, and buildings within each ring.'
```

```
return [summary, artifact_name, artifact_description, map_filename]
```